

Introduction to the semantics of modal logic

Lectures 1–4

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Structures and languages

Structures

A set of points, possibly with a relation between points and subsets:

- a point belongs to a subset;
- sees it, or can reach it;
- is close to it.

Formal language

An *alphabet* supplies symbols.

A *grammar* tells us how to combine them into *formulas*.

Interpretation connects a language with a structure

⇓ a **model**: truth of formulas at points

Truth and validity let the language describe features of structures.

Four lectures: sets, relations and spaces

- | | | |
|---|----------------------------|-------------------------------|
| 1 | Propositional logic | Sets and truth-sets |
| 2 | Kripke semantics | Seeing and reaching |
| 3 | Topological semantics | Closeness and closure |
| 4 | Admissible spatial regions | Chequered and polyhedral sets |

How much can a language describe?

When do proofs capture semantic truth?

When can a finite model represent what matters?

A query

“Find pages about linguistics or logic that also mention funding.” As a Boolean web-search query:

`(linguistics OR logic) AND funding`

OR: either topic, including pages mentioning both.

AND: the funding condition must also hold.

The answer is a set of matching pages.

We use an idealized search index and leave ranking aside.

Give the query a symbolic form

ℓ mentions linguistics

g mentions logic

f mentions funding

(linguistics OR logic) AND funding

$\Downarrow$

$$(\ell \vee g) \wedge f$$

$\vee$ means OR; $\wedge$ means AND.

Which pages match?

Our fictional search index		ℓ	g	f
A	Language and Logic Summer School	T	T	T
B	Introduction to Linguistics	T	F	F
C	Logic and Computation Summer School	F	T	T
D	Language, Logic and Cognition Workshop	T	T	F
E	Scholarships for Linguistics Students	T	F	T
F	Student Accommodation Guide	F	F	F

The T/F entries describe what each page contains, not just what its title says.

$$\llbracket (\ell \vee g) \wedge f \rrbracket = \{A, C, E\}. \quad \llbracket \ell \wedge \neg g \rrbracket = \{B, E\}.$$

The same idea works in a museum



$$X = \{E, C, W, S, Y\}.$$

Our objects are **five locations**, each treated as one element of X .

A model interprets the letters

Structure: a nonempty set X .

Language: letters $p, s, \dots$ and connectives.

Valuation: V assigns each letter a subset of X .

p : contains paintings

$$V(p) = \{E, C, W\}$$

s : contains sculptures

$$V(s) = \{C, Y\}$$

The model is the pair $\mathcal{M} = (X, V)$.

One structure, different interpretations

Add a letter o : “is open to visitors”.

All locations open $V(o) = X$

West Hall closes $V'(o) = X \setminus \{W\}$

The locations are the same. Their interpretation has changed.

$$(X, V) \neq (X, V').$$

A structure does not determine a unique model.

Formulas are built from smaller formulas

$p, s, o, \dots$ propositional letters

$\neg\varphi$ not φ

$\varphi \wedge \psi$ φ and ψ

$\varphi \vee \psi$ φ or ψ (or both)

$\varphi \rightarrow \psi$ if φ , then ψ

Start with letters; apply these rules finitely many times.

$$s \longrightarrow \neg s \longrightarrow p \wedge \neg s.$$

Truth at a point means set membership

$$\mathcal{M}, x \models \varphi \iff x \in \llbracket \varphi \rrbracket_{\mathcal{M}}$$

Read: “ φ is true at x in model $\mathcal{M}$.”

For a letter, $\llbracket p \rrbracket_{\mathcal{M}} = V(p)$.

$$\mathcal{M}, E \models p \quad \text{but} \quad \mathcal{M}, Y \not\models p.$$

How do we find the truth-set of a compound formula?

Connectives become operations on sets

not $\llbracket \neg \varphi \rrbracket = X \setminus \llbracket \varphi \rrbracket$ complement

and $\llbracket \varphi \wedge \psi \rrbracket = \llbracket \varphi \rrbracket \cap \llbracket \psi \rrbracket$ intersection

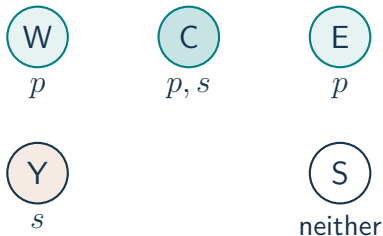
or $\llbracket \varphi \vee \psi \rrbracket = \llbracket \varphi \rrbracket \cup \llbracket \psi \rrbracket$ union

$$\llbracket p \wedge s \rrbracket = \{C\}, \quad \llbracket p \vee s \rrbracket = \{E, C, W, Y\}.$$

Interpret the letters, then build the answer step by step.

Regroup the same five locations

Museum layout



Grouped by yes/no answers

$p \wedge \neg s$ E, W	$p \wedge s$ C
$\neg p \wedge \neg s$ S	$\neg p \wedge s$ Y

Same model: only the drawing has changed.

First split by p ; within each part, split by s . E and W stay together.

Your turn: describe the locations

$p \wedge \neg s$ E, W	$p \wedge s$ C
$\neg p \wedge \neg s$ S	$\neg p \wedge s$ Y

Find the truth-sets:

$$\neg(p \vee s), \quad p \vee \neg p.$$

Find a formula describing:

$$\{C, Y\}, \quad \{E, W, Y\}.$$

Can you describe East Hall alone using only p and s ?

Implication excludes one combination

$\varphi \rightarrow \psi$ means $\neg\varphi \vee \psi$.

φ	ψ	$\varphi \rightarrow \psi$
T	T	T
T	F	F
F	T	T
F	F	T

“If it contains paintings, it is open.”

$$\llbracket p \rightarrow o \rrbracket = (X \setminus \llbracket p \rrbracket) \cup \llbracket o \rrbracket.$$

It fails precisely at a location with paintings that is closed.

Brackets and the universe both matter

Brackets: these are different search criteria.

$$(\ell \vee g) \wedge f \quad \text{and} \quad \ell \vee (g \wedge f).$$

A linguistics page with no funding satisfies only the second.

Universe: “not about linguistics” means

$$X \setminus \llbracket \ell \rrbracket.$$

It includes only pages in our chosen index X .

True here, or true everywhere?

At a point $\mathcal{M}, x \models \varphi$ $x \in \llbracket \varphi \rrbracket_{\mathcal{M}}$

In a model $\mathcal{M} \models \varphi$ $\llbracket \varphi \rrbracket_{\mathcal{M}} = X$

On a structure $X \models \varphi$ every valuation on X

Valid in CPL $\models_{\text{CPL}} \varphi$ every nonempty X , every V

Each step asks for truth in more cases.

A model truth need not be a logical truth

In our all-open museum:

$$\mathcal{M} \models o \quad \text{and} \quad \mathcal{M} \models p \rightarrow o.$$

Close West Hall: both fail there in the new model.

But under *every* interpretation:

$$\llbracket p \vee \neg p \rrbracket = V(p) \cup (X \setminus V(p)) = X.$$

$p \vee \neg p$ **is valid. The letter o is not.**

Satisfiability asks for one witness

A formula is **satisfiable** if it is true at some point in some model.

$p \wedge s$ is satisfiable: Central Hall is a witness.

$p \wedge \neg p$ is unsatisfiable: its truth-set is always $\emptyset$.

φ is valid exactly when $\neg\varphi$ is unsatisfiable.

Consequence means preserving truth

$$p, p \rightarrow o \models o.$$

At any point in any model: if both premises are true, so is the conclusion.

Within a fixed model:

$$\mathcal{M} \models \varphi \rightarrow \psi \iff \llbracket \varphi \rrbracket_{\mathcal{M}} \subseteq \llbracket \psi \rrbracket_{\mathcal{M}}.$$

A globally true implication expresses inclusion of truth-sets.

A valid implication indicates logical entailment.

Proofs manipulate formulas

A **Hilbert system** specifies axiom schemes and inference rules.

One axiom scheme:

$$\varphi \rightarrow (\psi \rightarrow \varphi).$$

One rule, **modus ponens**:

$$\frac{\varphi \quad \varphi \rightarrow \psi}{\psi}.$$

A proof is a finite list: each line is an axiom instance or follows by a rule.

The axiom above is only an example, not a complete basis. One possible full axiomatic basis is given in an appendix.

Soundness and completeness

$$\underbrace{\vdash_{\text{CPL}} \varphi}_{\text{there is a proof}} \iff \underbrace{\models_{\text{CPL}} \varphi}_{\text{true in every model}} .$$

Soundness: provable formulas are valid.

Completeness: valid formulas are provable.

Two different ways of describing the same logical truths.

For CPL, one point is enough

Suppose φ is false at x in a model (X, V) .

On the singleton $\{*\}$, copy the truth-values at x :

$$V_*(p) = \begin{cases} \{*\}, & x \in V(p), \\ \emptyset, & x \notin V(p). \end{cases}$$

Every formula has the same truth-value at $*$ as at x .

Every nonvalid CPL formula has a one-point countermodel.

Likewise, every satisfiable formula has a singleton model: the **finite model property**.

One structure, all its valuations

$$\models_{\text{CPL}} \varphi \iff \{*\} \models \varphi.$$

We must still test **every valuation** on that singleton.

For n letters, there are 2^n assignments: the rows of a truth table.

All nonempty sets validate exactly the same CPL formulas.

CPL validity cannot tell a one-element set from an infinite set.

A tiny model can still leave a hard search

One point suffices, but we must choose its truth-values.

30 letters $\implies 2^{30} > 1$ billion assignments.

Checking a given assignment is easy. Finding one that works can be demanding.
Is there an algorithm deciding satisfiability for every formula in polynomial time?

This is equivalent to $P = NP$: a Millennium Prize Problem.

Clay Mathematics Institute: P vs NP

Which subsets can formulas describe?

$p \wedge \neg s$ E, W	$p \wedge s$ C
$\neg p \wedge \neg s$ S	$\neg p \wedge s$ Y

A subset $A \subseteq X$ is **definable** using p, s if

$$A = \llbracket \varphi \rrbracket$$

for some formula using only p, s .

$\{C\}$ is definable: $p \wedge s$.

$\{E\}$ is not: E and W agree on both letters.

Definable sets are unions of these groups

$p \wedge \neg s$ E, W	$p \wedge s$ C
$\neg p \wedge \neg s$ S	$\neg p \wedge s$ Y

Choose any of the four groups and take their union.

There are $2^4 = 16$ choices, including $\emptyset$ and X .

These are *exactly* the truth-sets of formulas using p, s .

Infinitely many formulas, only sixteen different truth-sets.

Finitely many letters, finitely many sets

Fix n letters in any model, even an infinite one.

Group points with the same yes/no answers to those letters.

At most 2^n groups; at most 2^{2^n} unions.

These unions form the **Boolean algebra generated** by the chosen truth-sets: closed under complement, union and intersection.

Every formula using these letters defines one of these unions.

Two reasons CPL is manageable

To test validity: one singleton structure is enough, with all valuations.

Inside a fixed model: finitely many letters generate only finitely many truth-sets.

Both reflect the same fact:

CPL only asks which properties hold at the current point.

What if we ask about another room?

Let m mean “contains merchandise”; it is true only at the shop.

East and West agree on p, s, m . Suppose East Hall has a direct passage to the shop, while West Hall does not.

$\Diamond m$: “I can move directly to a place with merchandise.”

This is true at East Hall and false at West Hall.

Next lecture: add a relation, and look beyond here.

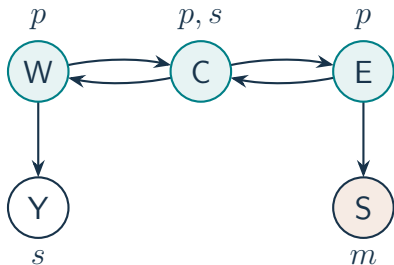
Introduction to the semantics of modal logic

Lecture 2: What can a point see?

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The same museum, now with doors



p : contains paintings

s : contains sculptures

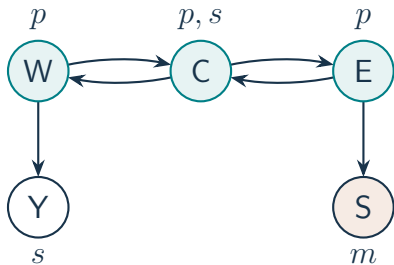
m : contains merchandise

Hall doors work both ways.

East leads to the shop; West leads to the courtyard. These passages are one-way.

Only the displayed arrows count. A move means one direct passage.

A relation turns the set into a frame



A Kripke frame is $\mathcal{F} = (X, R)$.
 xRy means an arrow from x to y .
A valuation gives a model:

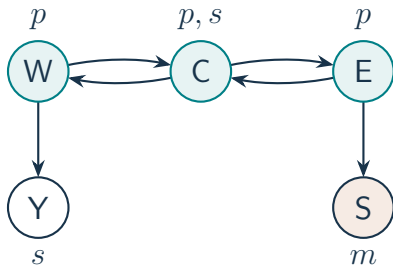
$$\mathcal{M} = (X, R, V).$$

The frame supplies the arrows; the valuation supplies the properties.

A new connective: diamond

“I can move directly to a place with merchandise.”

$\Diamond m$



True at East Hall.
False at West Hall.
Yet E and W agree on p, s, m .

Modal formulas can distinguish points that CPL could not.

Possibility means seeing a truth-set

$$\mathcal{M}, x \models \Diamond\varphi \iff \text{some } y \text{ with } xRy \text{ belongs to } \llbracket\varphi\rrbracket.$$

Define an operation on subsets:

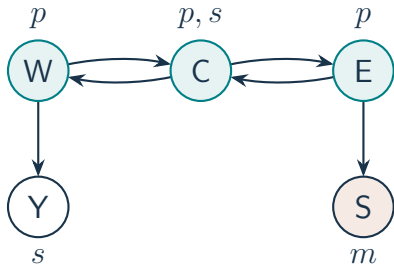
$$D_R(A) = \{x \in X : \text{for some } y \in A, xRy\}.$$

Then

$$\boxed{\llbracket\Diamond\varphi\rrbracket = D_R(\llbracket\varphi\rrbracket).}$$

Start with the target set; collect the points that can see it.

Work backwards from the target set



$$\llbracket m \rrbracket = \{S\}.$$

Which locations have an arrow into this set?

$$\llbracket \Diamond m \rrbracket = \{E\}.$$

$$\llbracket \Diamond \Diamond m \rrbracket = \{C\}.$$

Each diamond adds one step, working from the inside out.

Necessity means all accessible points qualify

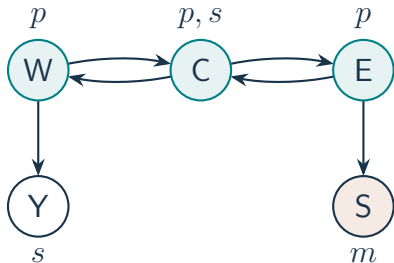
$$\Box\varphi := \neg\Diamond\neg\varphi.$$

“There is no accessible point where φ fails.”

$$\mathcal{M}, x \models \Box\varphi \iff \text{every } y \text{ with } xRy \text{ belongs to } \llbracket\varphi\rrbracket.$$

$$\llbracket\Box\varphi\rrbracket = X \setminus D_R(X \setminus \llbracket\varphi\rrbracket).$$

Some successors, or all successors?



$$\llbracket p \rrbracket = \{E, C, W\}.$$

$$\llbracket \Diamond p \rrbracket = \{E, C, W\}.$$

$$\llbracket \Box p \rrbracket = \{C, S, Y\}.$$

East sees paintings, but also the shop.
Why do S and Y satisfy $\Box p$?

At a dead end, every box is true

The shop and courtyard have no outgoing arrows in this model.

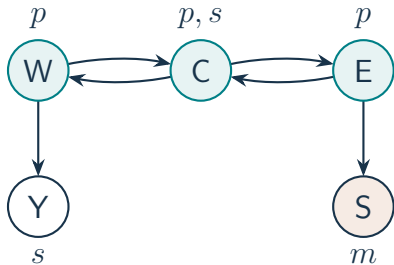
They cannot see a counterexample to any formula.

$$\mathcal{M}, S \models \Box\varphi, \quad \mathcal{M}, S \not\models \Diamond\varphi.$$

In particular, $\Box p$ is true there although p is false.

“All successors qualify” does not assert that any successors exist.

Your turn: follow the arrows



At Central Hall, which are true?

- | | |
|-------------------------|-------|
| 1. $\Diamond m$ | false |
| 2. $\Diamond\Diamond m$ | true |
| 3. $\Box p$ | true |
| 4. $\Box\Diamond p$ | true |

For the last formula: E and W can each move to C, which has paintings.

Truth in a model is not validity on a frame

$\mathcal{M}, x \models \varphi$ at this point, with this valuation

$\mathcal{M} \models \varphi$ at every point, with this valuation

$\mathcal{F} \models \varphi$ at every point, under every valuation

$\models \varphi$ on every frame

Frame validity tests the arrows by trying every colouring.

K: what holds for every accessibility relation?

Start with all classical tautologies, allowing modal formulas in place of variables.

Add all instances of

$$\boxed{\Box(\varphi \rightarrow \psi) \rightarrow (\Box\varphi \rightarrow \Box\psi).}$$

Use modus ponens and **necessitation**:

$$\frac{\vdash \varphi}{\vdash \Box\varphi}.$$

The formulas provable this way form the logic K.

Why the K axiom is sound

Suppose every successor satisfies $\varphi \rightarrow \psi$, and every successor satisfies φ .
Then every successor satisfies ψ .

$$\Box(\varphi \rightarrow \psi) \wedge \Box\varphi \implies \Box\psi.$$

In truth-set language, another universal law is

$$D_R(A \cup B) = D_R(A) \cup D_R(B).$$

Seeing an A -or- B point means seeing an A point or a B point.

A modal formula can define reflexivity

A relation is **reflexive** when every point sees itself. Can this be expressed by a validity of a single modal formula?

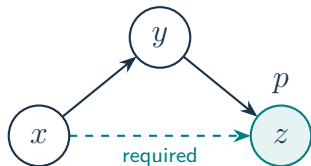
$$\boxed{\mathcal{F} \models p \rightarrow \Diamond p \iff R \text{ is reflexive.}}$$

If xRx , then whenever x has p , it can see a p -point: itself.

If x does not see itself, choose $V(p) = \{x\}$. Then $p \rightarrow \Diamond p$ fails at x .

One carefully chosen truth-set exposes a missing loop.

A modal formula can define transitivity



Transitive: if xRy and yRz , then xRz .

Which modal formula expresses this property?

Every two-step connection is also a one-step connection.

$$\mathcal{F} \models \Diamond\Diamond p \rightarrow \Diamond p \iff R \text{ is transitive.}$$

Direct passage and reachability differ

Keep the museum, but change what counts as accessibility.

R : direct passage one step

R^* : reachability zero or more steps

With R^* , every location reaches itself. Routes can be joined together.

R^* is reflexive and transitive.

With R , $\Diamond m$ holds only at E. With R^* , it holds at E, C, W and S.

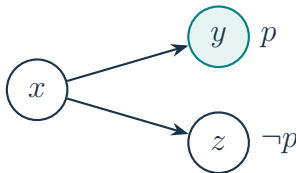
K, K4 and S4: axioms and intended frames

Logic	Add to K	Frames
K	nothing	arbitrary
K4	$\Diamond\Diamond p \rightarrow \Diamond p$	transitive
S4	also $p \rightarrow \Diamond p$	reflexive, transitive

Soundness and completeness: in each row, provability is exactly validity on the indicated frames. Reachability in zero or more steps is reflexive and transitive.

One-point testing no longer suffices

Every singleton frame validates $\Diamond p \rightarrow \Box p$, with or without a loop.

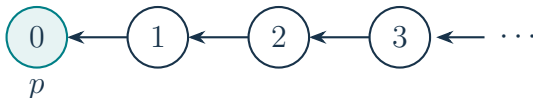


At x , $\Diamond p$ is true and $\Box p$ is false.

Can we still find a finite countermodel whenever we need one?

Modal operations create new truth-sets

Take $X = \{0, 1, 2, \dots\}$, with $nR(n-1)$ for $n \geq 1$, and $V(p) = \{0\}$.



$$\llbracket \Diamond p \rrbracket = \{1\}, \quad \llbracket \Diamond \Diamond p \rrbracket = \{2\}, \quad \dots$$

One variable now yields infinitely many different truth-sets.

This is a direct-step relation; it is not transitive.

A fixed formula still has only finitely many subformulas.

Can we keep only a finite record?

A model may have infinitely many points and infinitely many truth-sets.

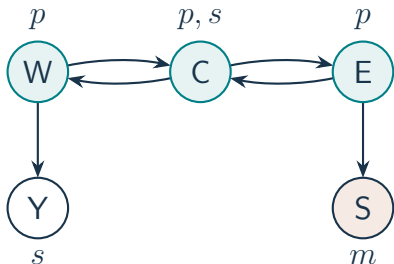
But a fixed formula φ has only finitely many subformulas.

Suppose we remember the truth-values of just those subformulas, and merge points that agree on all of them.

Can we choose arrows between the groups so that these answers survive?

This construction is called **filtration**.

Filtration: remember the relevant questions



Suppose we want to evaluate

$$\varphi = \Diamond\Diamond m.$$

Its subformulas are

$$m, \quad \Diamond m, \quad \Diamond\Diamond m.$$

Remember their truth-values at each point.

Merge points only when they agree on every question in this list.

Only m occurs in our formula; we need not record p or s .

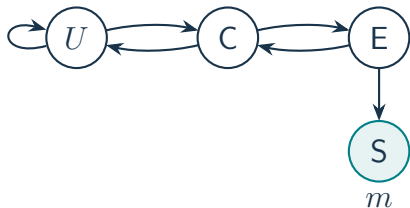
The record tells us what can be merged

Location	m	$\Diamond m$	$\Diamond\Diamond m$
E	F	T	F
C	F	F	T
W	F	F	F
S	T	F	F
Y	F	F	F

W and Y agree on this list. Merge them into one point U .

The four new points are $\{E\}, \{C\}, \{S\}, \{W, Y\}$.

Keep an arrow if an old arrow witnesses it



$U = \{W, Y\}$.

$W \rightarrow Y$ becomes a loop at U .

The three recorded truth-values are unchanged.

Selected truths survive; other truths may change.

Y originally had no successor; its image U does. So $\Diamond \top$ is not preserved.

Filtration makes the record finite

Fix the finite set Σ of subformulas of φ .

Group points that agree on every formula in Σ .

At most $2^{|\Sigma|}$ groups, even in an infinite model.

Filtration lemma: with the appropriate quotient relation and valuation, every formula in Σ keeps its truth-value.

For K, use the witnessed-arrow relation of our example.

Finite countermodels suffice for all three logics

Finite model property for K, K4 and S4:

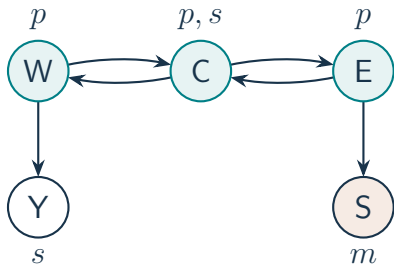
If a formula is not provable, it fails in some finite model of the appropriate kind.

Completeness gives a countermodel; filtration makes it finite.

For K4 and S4, filtration must also preserve transitivity. We adjust the quotient relation accordingly.

Finite does not mean one fixed size, or computationally easy.

Keep the successors when restricting a model



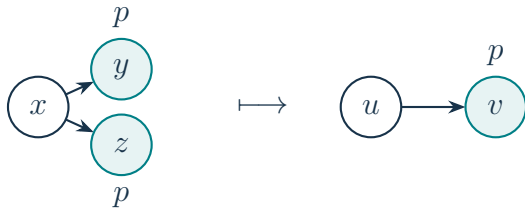
Keep only E: $\Diamond m$ changes from true to false there.

Keep S alone: no possible moves are lost.

A **generated submodel** keeps every successor of every retained point.

These restrictions preserve every formula, not just a chosen finite list.

Merging points must respect their successors



Merge y, z into v , and send x to u .

- Basic truths agree at corresponding points.
- Arrows are preserved; every move from an image can be matched at the original.

A **bounded morphism** preserves every modal formula.

Validity survives certain operations

Frame validity is preserved when we:

- take a nonempty generated subframe;
- take a surjective bounded-morphic image;
- form a disjoint union of frames all validating the formula.

A class defined by modal validity must be closed under these operations.

For example, “has exactly one point” is not definable: two disjoint copies still satisfy every formula valid on the original frame.

Reflexive, transitive seeing obeys closure laws

Return to a reflexive, transitive frame. Recall

$$D_R(A) = \{x : x \text{ sees some point of } A\}.$$

$$\begin{aligned} D_R(\emptyset) &= \emptyset, & D_R(A \cup B) &= D_R(A) \cup D_R(B), \\ A &\subseteq D_R(A), & D_R(D_R(A)) &= D_R(A). \end{aligned}$$

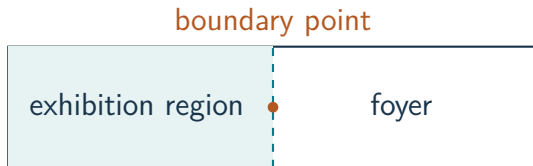
These are the laws of a **topological closure operation**.

Next: interpret diamond as closeness in a geometric space.

Lecture 3: from rooms to regions

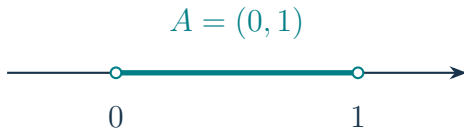
In Kripke semantics, each hall was a single point.

Now points are actual positions, and propositional letters describe **regions** of the floor.



Diamond now means “close to this region”.

Close means every neighbourhood meets the set



0 is not in $(0, 1)$, but every open interval around 0 meets it.

A point is in $\text{cl}(A)$ if every neighbourhood of it meets A .

$$\text{cl}((0, 1)) = [0, 1].$$

There is no smallest permitted distance: every scale matters.

Diamond is closure; box is interior

$$\llbracket \Diamond \varphi \rrbracket = \text{cl}(\llbracket \varphi \rrbracket)$$

Boolean connectives still mean complement, intersection and union.

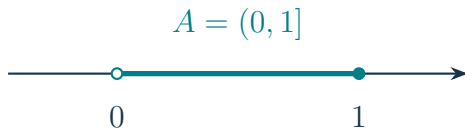
$$\llbracket \Box \varphi \rrbracket = X \setminus \text{cl}(X \setminus \llbracket \varphi \rrbracket) = \text{int}(\llbracket \varphi \rrbracket).$$

Interior: points with a whole neighbourhood inside the set.

A topological model is a space with an interpretation of letters as subsets.

Your turn: an interval as a truth-set

Work in the real line, with $\llbracket p \rrbracket = (0, 1]$.



$$\llbracket \Diamond p \rrbracket = [0, 1],$$

$$\llbracket \Box p \rrbracket = (0, 1),$$

$$\llbracket \Diamond p \wedge \neg p \rrbracket = \{0\}.$$

A useful operation: the outside boundary

$$B\varphi := \Diamond\varphi \wedge \neg\varphi.$$

$$\llbracket B\varphi \rrbracket = \text{cl}(\llbracket \varphi \rrbracket) \setminus \llbracket \varphi \rrbracket.$$

These are points outside the region but close to it.

For $(0, 1)$, the outside boundary is $\{0, 1\}$.

For $[0, 1]$, it is empty.

Boundary membership matters: outside boundary is not the full boundary.

Closeness concerns the whole subset

In $\mathbb{R}$, $0 \in \text{cl}((0, 1))$.

But for every individual $y \in (0, 1)$,

$$0 \notin \text{cl}(\{y\}) = \{y\}.$$

So ordinary spatial closure cannot be recovered from a relation between individual points in the Kripke way.

Closeness to a region need not mean closeness to one of its individual points.

S4 is exactly the logic of topological closure

Soundness: every theorem of S4 is valid in every topological space.

Completeness: every formula not provable in S4 fails in a topological model.

In fact, the ordinary real line already suffices, when every subset is allowed as a valuation.

Next lecture: what changes if we allow only well-behaved regions?

Lecture 4: geometry with a finite description

Our geometric spaces have infinitely many points.

What if we allow only regions with a finite geometric description?

Can a finite diagram record everything our formulas can say?

$$\llbracket \Diamond \varphi \rrbracket = \text{cl}(\llbracket \varphi \rrbracket).$$

Choose simple regions, divide the space into cells, and record their faces.

Allow regions with a finite geometric description

On the line: finite unions of intervals and individual points.

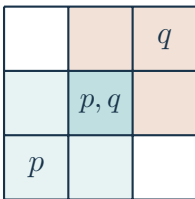
In the plane: finite unions of axis-aligned rectangles, segments and points.

Endpoints and edges may be included or excluded.

These are chequered subsets.

Complement, finite union and closure keep us within this collection.

A finite grid records the relevant boundaries



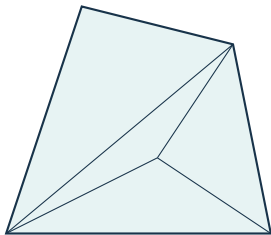
Use all the endpoints of the finitely many chosen rectangles.

This gives a common grid.

Keep its open cells, edge pieces and vertices separately.

Formula evaluation never needs to cut one of these cells in two.

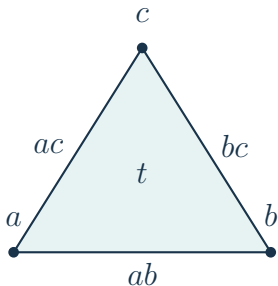
Triangles allow more flexible regions



Divide a polygon into triangles.
A compatible **triangulation** respects the boundaries of all chosen regions.
A **polyhedral region** may include interiors, edges and vertices.

Boolean combinations of subpolyhedra give the allowed subsets.

A triangle has seven cells

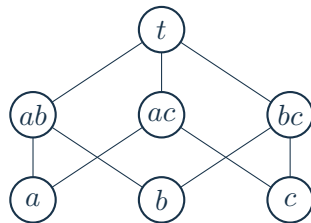
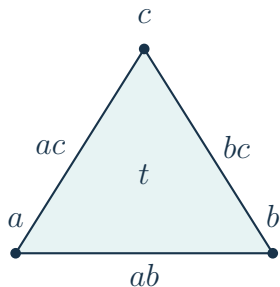


One open triangle: t .
Three open edges: ab, ac, bc .
Three vertices: a, b, c .

The seven cells are disjoint and cover the whole closed triangle.

“Open edge” means the segment without its endpoints.

Record which cells are faces of which

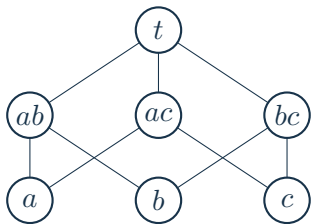


Read the order upwards: a vertex sees its incident edges and triangle.

uRv means that u is a face of v , including $u = v$.

The diagram omits loops and arrows implied by transitivity.

Closure becomes a finite calculation



Let p hold only on the open edge ab .

$$\llbracket p \rrbracket_F = \{ab\}.$$

$$\llbracket \Diamond p \rrbracket_F = \{a, ab, b\}.$$

In the triangle, this is the closed edge $[a, b]$.

To take closure, add all faces of the selected cells.

The cell map preserves every modal formula

Send each spatial point to the unique cell containing it:

$$f : P \longrightarrow F.$$

If the chosen propositional regions are unions of cells, then

$$\boxed{\llbracket \varphi \rrbracket_P = f^{-1}(\llbracket \varphi \rrbracket_F)}$$

for every formula using those letters.

A finite poset faithfully represents this spatial model.

Each node represents a whole cell, often containing infinitely many points.

Why the finite description lasts

Start with finitely many polyhedral truth-sets.

Choose a common finite triangulation respecting them all.

Boolean operations and closure only select unions of its cells.

N cells $\implies$ at most 2^N truth-sets.

The finite-algebra phenomenon from CPL returns, now including diamond.

Can formulas detect dimension?

Recall $B\varphi = \Diamond\varphi \wedge \neg\varphi$:

$$\llbracket B\varphi \rrbracket = \text{cl}(\llbracket \varphi \rrbracket) \setminus \llbracket \varphi \rrbracket.$$

On the line, an allowed region is a finite union of intervals and points.

Its outside boundary is a finite set, hence closed. Taking its outside boundary again gives nothing:

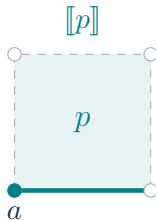
$$\boxed{\llbracket BBp \rrbracket = \emptyset.}$$

So $\neg BBp$ is valid on the line with these allowed regions.

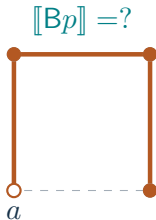
In the plane: take the outside boundary twice

Let $a = (0, 0)$ and work in $\mathbb{R}^2$ (or in the closed unit square).

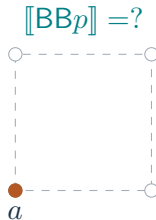
$$\llbracket p \rrbracket = (0, 1)^2 \cup ((0, 1) \times \{0\}) \cup \{a\}.$$



Open square, open bottom edge,
and vertex a .



The other three edges,
with a removed.



Only a is missing from
the closure of $\llbracket Bp \rrbracket$.

Solid edges / filled dots: included. Dashed edges / hollow dots: excluded.

$$\llbracket BBp \rrbracket = \{a\} \neq \emptyset.$$

The regions help determine the logic

With **all subsets**, the real line has logic S4.

With **polyhedral subsets**, additional formulas become valid; dimension can matter.

Finite face posets explain both the new laws and the finite representations.

Both the space and the allowed interpretations matter.

Application: compute where a spatial property holds

For example: “Which positions lie outside a restricted region but are close to it?”

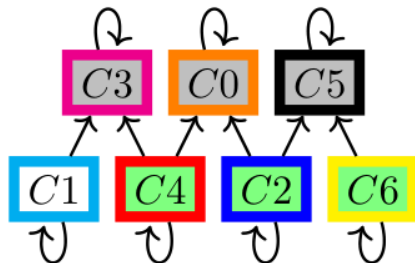
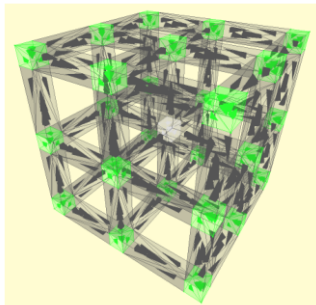
$$\llbracket \neg r \wedge \Diamond r \rrbracket.$$

spatial mesh $\longrightarrow$ finite cell model
 $\longrightarrow$ evaluate formula $\longrightarrow$ colour the answer

PolyLogicA implements geometric model checking. Richer spatial languages add path and reachability operators.

A formula becomes a query returning a region of space.

Application: simplify the finite representation



2,619 cells $\longrightarrow$ 7 states

Merge states indistinguishable by the chosen spatial language.

Bezhanishvili et al. (2026), Fig. 10; equivalence for $SLCS_{\eta}$.

One idea through the whole tutorial

A formula describes a truth-set.

CPL	combine subsets
Kripke semantics	see a subset through a relation
Topological semantics	be close to a subset
Polyhedral semantics	compute with finitely many cells

The language describes space; the finite model makes that description usable.

Appendix: a complete Hilbert basis

Take $\neg, \rightarrow$ as primitive connectives. Use all instances of:

$$\text{A1: } \varphi \rightarrow (\psi \rightarrow \varphi),$$

$$\text{A2: } (\varphi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \chi)),$$

$$\text{A3: } (\neg\psi \rightarrow \neg\varphi) \rightarrow (\varphi \rightarrow \psi).$$

Rule: modus ponens.

$$\varphi \vee \psi := \neg\varphi \rightarrow \psi, \quad \varphi \wedge \psi := \neg(\varphi \rightarrow \neg\psi).$$

$\top$ abbreviates $p \rightarrow p$; $\perp$ abbreviates $\neg\top$.

Appendix: a small formal proof

A proof of $p \rightarrow p$:

- | | |
|---|----------|
| 1. $p \rightarrow ((p \rightarrow p) \rightarrow p)$ | A1 |
| 2. $p \rightarrow (p \rightarrow p)$ | A1 |
| 3. $(p \rightarrow ((p \rightarrow p) \rightarrow p)) \rightarrow$
$((p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p))$ | A2 |
| 4. $(p \rightarrow (p \rightarrow p)) \rightarrow (p \rightarrow p)$ | MP: 1, 3 |
| 5. $p \rightarrow p$ | MP: 2, 4 |

Even an obvious truth can have a fully explicit formal proof.

Appendix: theories and finite models

For any set of premises Γ :

$$\Gamma \vdash_{\text{CPL}} \varphi \iff \Gamma \models \varphi.$$

Equivalently, every syntactically consistent set of formulas has a model.

Finite model property: every satisfiable CPL formula has a finite model—indeed, a singleton model.

$\Gamma \models \varphi$ means: at every point satisfying all of Γ , φ is also true.

Appendix: definability of structures

A class $\mathcal{C}$ of nonempty sets is defined by formulas Σ if

$$X \in \mathcal{C} \iff X \models \sigma \text{ for every } \sigma \in \Sigma.$$

For CPL, such a class is either all nonempty sets or none.

For example, “has at least two elements” is not definable this way.

Defining structures differs from defining a subset in a model.

Appendix: pulling an interpretation back

Let $f : X \rightarrow Y$ and let V_Y interpret letters on Y . Define $V_X(p) = f^{-1}(V_Y(p))$.
Then

$$\llbracket \varphi \rrbracket_X = f^{-1}(\llbracket \varphi \rrbracket_Y)$$

for every CPL formula φ .

So φ is true at x exactly when it is true at $f(x)$.

Inverse images preserve complement, intersection and union. This explains restriction to a point and quotienting by truth-patterns.

Diamond preserves unions, but not intersections

$$\Diamond(p \vee q) \leftrightarrow (\Diamond p \vee \Diamond q)$$

is valid on every frame.

But

$$(\Diamond p \wedge \Diamond q) \rightarrow \Diamond(p \wedge q)$$

is not: a point may see a p -only point and a q -only point.

Two existential claims may have different witnesses.

The formal bounded-morphism conditions

A map $f : (X, R) \rightarrow (Y, S)$ is a **bounded morphism** if:

1. xRx' implies $f(x)Sf(x')$;
2. if $f(x)Sy$, some x' satisfies xRx' and $f(x') = y$.

With valuations related by $V_X(p) = f^{-1}(V_Y(p))$,

$$\mathcal{M}_X, x \models \varphi \iff \mathcal{M}_Y, f(x) \models \varphi.$$

If f is onto, validity on (X, R) passes to (Y, S) .

The box forms of T and 4

Over K, the following axiom *schemes* are equivalent:

	Diamond form	Box form
T	$p \rightarrow \Diamond p$	$\Box p \rightarrow p$
4	$\Diamond \Diamond p \rightarrow \Diamond p$	$\Box p \rightarrow \Box \Box p$

T expresses reflexivity; 4 expresses transitivity.

These are equivalences of axioms, not pointwise equivalences with the same valuation.

Reference: the filtration definition

Let Σ be the set of all subformulas of the formula being tested.

$$x \sim y \iff x, y \text{ agree on every formula in } \Sigma.$$

$$[x]R_{\min}[y] \iff x'Ry' \text{ for some } x' \sim x, y' \sim y.$$

For each propositional letter $p \in \Sigma$, put

$$V_f(p) = \{[x] : x \in V(p)\}.$$

Then, for $\psi \in \Sigma$,

$$\mathcal{M}, x \models \psi \iff \mathcal{M}_f, [x] \models \psi.$$

Reference: transitive filtration

If the original relation is transitive, use

$$R_f = (R_{\min})^+,$$

the transitive closure of the witnessed-arrow relation.

The filtration lemma still holds for every $\psi \in \Sigma$.

If the source is also reflexive, $R_{\min}$ and R_f are reflexive.

This gives finite countermodels of the right kind for K4 and S4.

Reference: truth-preserving maps

For Kripke frames, a bounded morphism preserves arrows and allows each move from an image to be matched at its original.

For topological spaces, a continuous open map $f : X \rightarrow Y$ satisfies

$$f^{-1}(\text{cl}_Y A) = \text{cl}_X(f^{-1}A).$$

With valuations pulled back along f , all modal truth is preserved and reflected at corresponding points.

If f is onto, validity on the source passes to the target.

Reference: open sets and finite spaces

A topology is a collection of open subsets containing $\emptyset, X$, closed under arbitrary unions and finite intersections.

In an S4 frame, declare U open when

$$x \in U \text{ and } xRy \implies y \in U.$$

Then $\text{cl } A = D_R(A)$.

On finite spaces, topology and reflexive transitive accessibility determine one another.

Reference: Grz and local finiteness

$$\text{Grz} = \text{S4} + \left[\Box(\Box(p \rightarrow \Box p) \rightarrow p) \rightarrow p \right].$$

Grz is complete for finite partial orders.

Polyhedral logics extend Grz. For a fixed finite-dimensional polyhedron, its PL logic is locally finite. Here logical local finiteness means finitely many inequivalent formulas in each fixed finite vocabulary.

Reference: general topological models

A general topological structure is $(X, \mathcal{A})$, where $\mathcal{A}$ is a Boolean algebra of subsets closed under topological closure.

Only valuations into $\mathcal{A}$ are allowed.

$$\text{Log}(X, \mathcal{A}) = \{\varphi : \varphi \text{ is valid for all allowed valuations}\}.$$

Polyhedral semantics makes a geometric choice of $\mathcal{A}$.

Reference: connected logics on the real line

A closure algebra is **connected** if its only elements that are both open and closed are $\emptyset$ and X .

A logic is connected if it is the logic of some connected closure algebra.

Representation theorem: for every connected normal extension L of $S4$, some admissible algebra $\mathcal{A}$ on $\mathbb{R}$ satisfies

$$L = \text{Log}(\mathbb{R}, \mathcal{A}).$$

Reference: more freedom in the admissible sets

Let $X = \bigcup_{n \geq 1} (n, n + 1) \subseteq \mathbb{R}$.

Every consistent normal extension L of S4 can be represented as

$$L = \text{Log}(X, \mathcal{A})$$

for a suitable closure algebra $\mathcal{A}$ of admissible subsets.

A simple underlying space can support complicated logics.

The freedom lies in the choice of admissible sets; they need not be polyhedral.

Reference: the map behind the representation

Every finite connected S4 frame, viewed as a finite space, is a continuous open image of the real line.

For such a map $f : \mathbb{R} \rightarrow F$, take

$$\mathcal{A} = \{f^{-1}(A) : A \subseteq F\}.$$

Then $\text{Log}(\mathbb{R}, \mathcal{A}) = \text{Log}(F)$.

The pullback of truth-sets links geometric and relational semantics.

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